PARSHWANATH CHARITABLE TRUST'S
A.P. SHAH INSTITUTE OF TECHNOLOGY
Department of Computer Science and Engineering
Data Science

Lecture Reminder and Timetable Management System

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Project Guide
~ Ms. Swati Parhad

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1. Introduction

- Problem Identified :
  - Students often do not know the correct classroom or lecture timing.
  - When lectures are cancelled or rescheduled, teachers find it difficult to inform all students quickly about lecture schedules or updates.
  - There is no centralized system where students can quickly check the latest lecture schedule.

1. Introduction

- Solution Proposed :
  - Develop a mobile application where teachers can upload and manage the lecture timetable easily.
  - The timetable is stored , allowing students and teachers to access the latest lecture schedule anytime.
  - The app sends automatic reminders / notifications to students and respective teacher before the lecture begins, helping them reach the correct classroom on time.

2. Objectives

1. To provide a centralized and structured system for creating, updating, and sharing lecture schedules between faculty and students.

2. To enable real-time notifications and reminders for scheduled lectures, including subject, time, and classroom details.

3. To facilitate instant communication of important updates such as lecture cancellations, substitutions, or rescheduling.

4. To ensure easy accessibility of timetable and notification information anytime and anywhere through mobile devices.

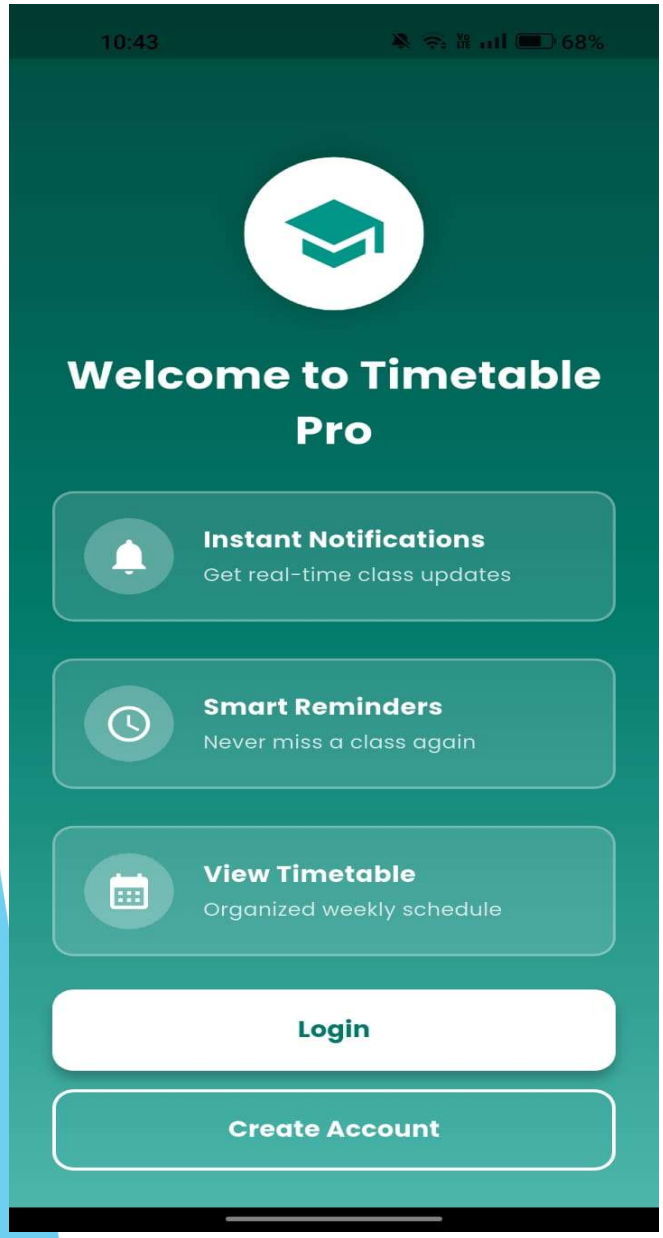
3. Scope

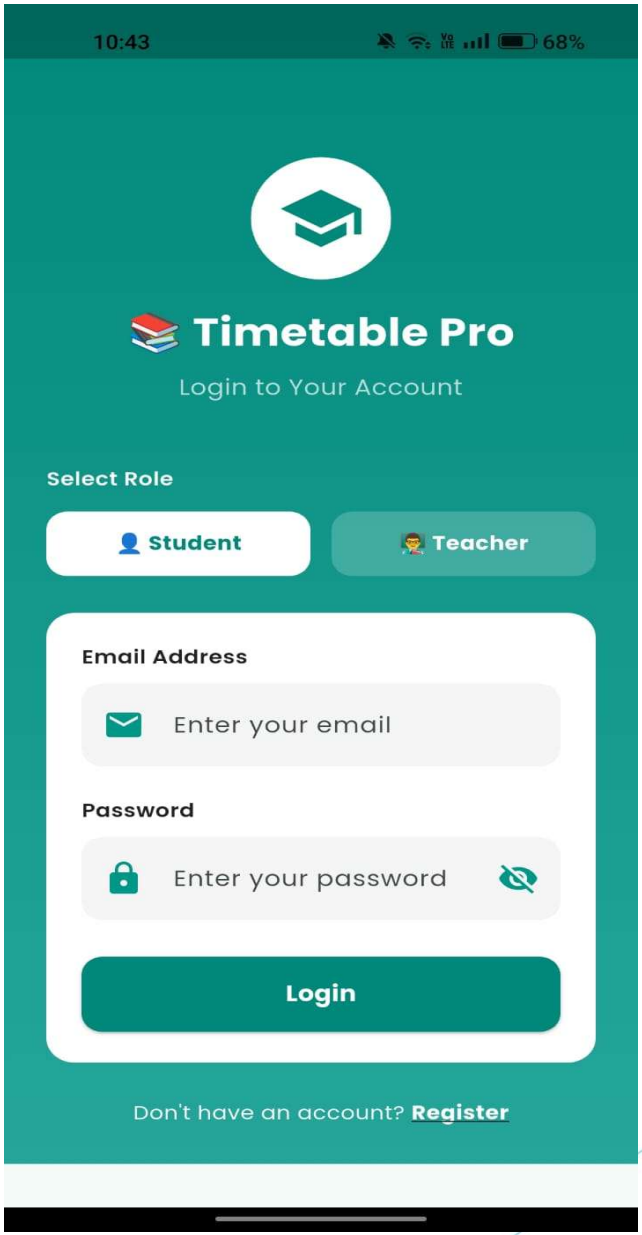
1. Can be used in educational departments or institutes to maintain organized lecture scheduling.
2. Can be useful to both students and teachers for easily accessing lecture schedules and receiving timely lecture reminders.

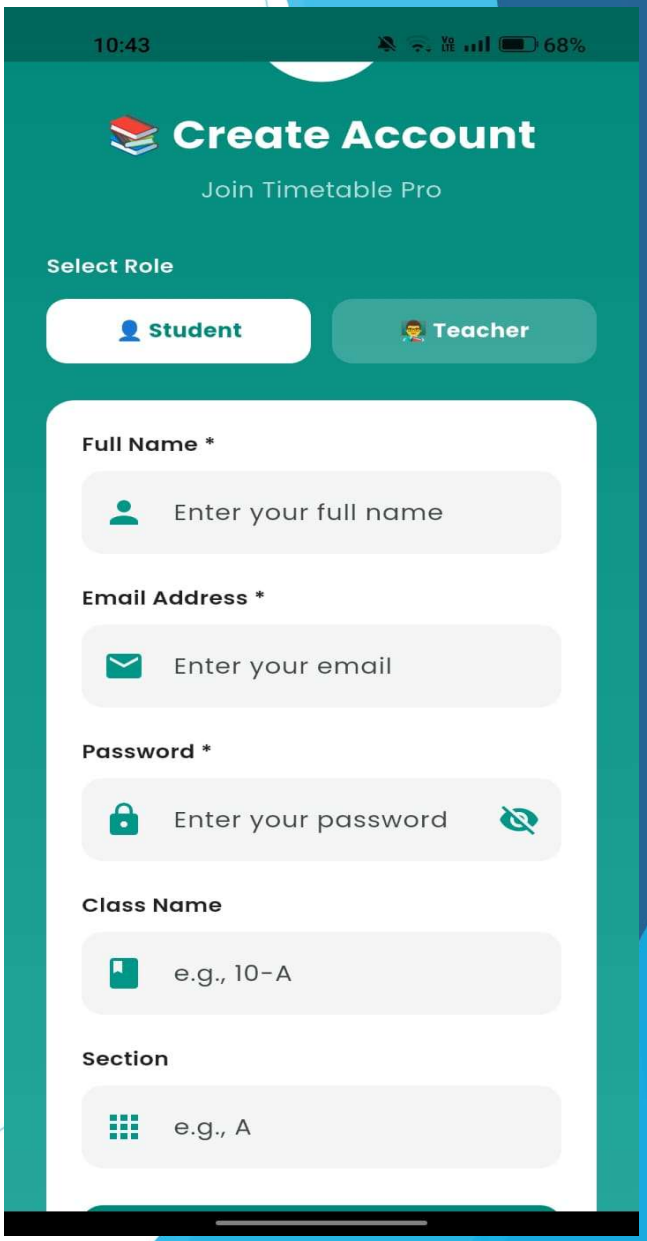
4. Feature /Functionality

1. Feature 1 : Teacher Timetable Upload
   - Teachers can upload and manage lecture timetable details such as subject name, lecture time, and classroom through the mobile application.
2. Feature 2 :Lecture Schedule Viewing
   - Students and teachers can view the lecture timetable anytime through the application.
3. Feature 3 :Lecture Reminder System
   - The app provides automatic reminders before the lecture begins. This helps students and teachers stay informed and reduces the chances of missing a lecture.

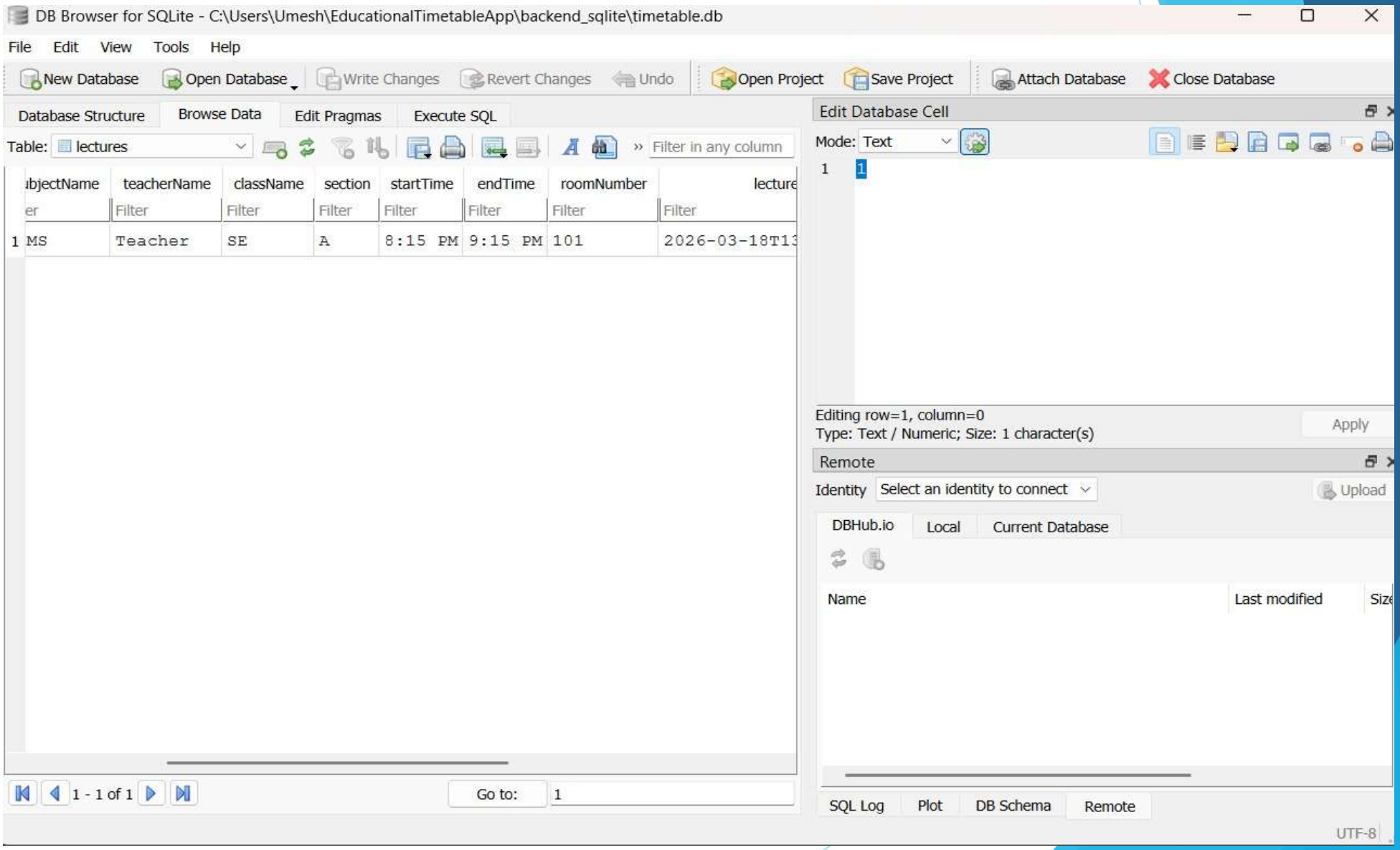
5. Outcome of Project







6. Backend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New Database
Open Database
Write Changes
Revert Changes
Undo
Open Project
Save Project
Attach Database
Close Database
Database Structure
Browse Data
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1 - 1 of 1
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Editing row=1, column=0
Type: Text / Numeric; Size: 1 character(s)
Apply
Remote
Identity
Select an identity to connect
Upload
DBHub.io
Local
Current Database
Name
Last modified
Size
SQL Log
Plot
DB Schema
Remote
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6. Backend

Umesh > EducationalTimetableApp > backend_sqlite >

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Sort View

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File Edit View

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    email TEXT UNIQUE NOT NULL,
    passwordHash TEXT NOT NULL,
    role TEXT NOT NULL,
    className TEXT,
    section TEXT,
    createdAt TEXT DEFAULT CURRENT_TIMESTAMP
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lectures (
    id TEXT PRIMARY KEY,
    subjectName TEXT NOT NULL,
    teacherName TEXT NOT NULL,
    className TEXT NOT NULL,
    section TEXT,
    startTime TEXT NOT NULL,
    endTime TEXT NOT NULL,
    roomNumber TEXT,
    lectureDate TEXT NOT NULL,
    isCancelled INTEGER DEFAULT 0,
    cancellationReason TEXT,
    createdAt TEXT DEFAULT CURRENT_TIMESTAMP
  )/□□□C□□ indexsqlite_autoindex_lectures_1lectures□,]□ □''□„ytablenotificationsnotifications□CREATE TABLE notifications (
    id TEXT PRIMARY KEY,
    studentId TEXT NOT NULL,
    lectureId TEXT,
    title TEXT NOT NULL,
    message TEXT NOT NULL,
```

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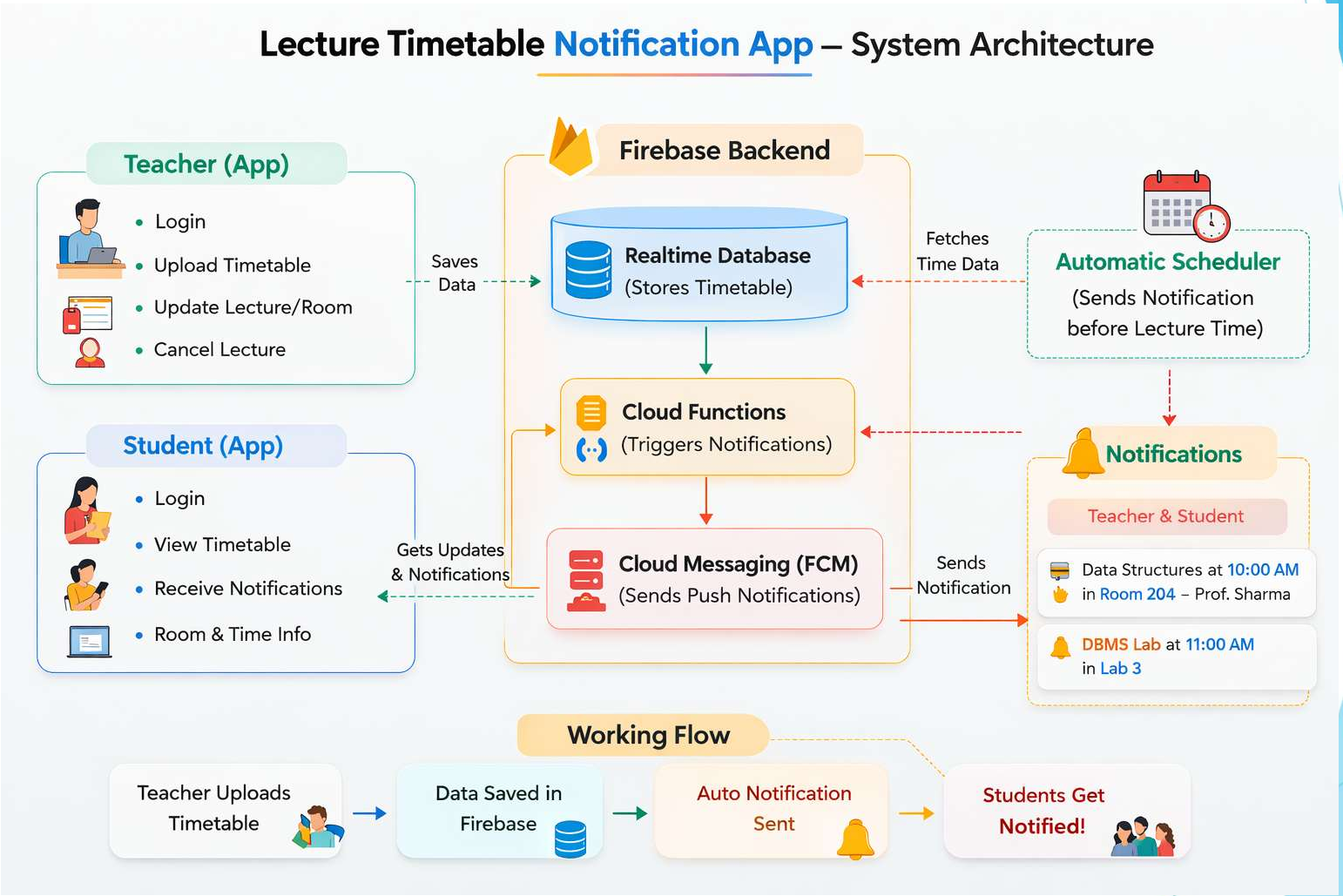
7. Technology Stack

1. Language: Java & Kotlin (Core logic).
2. IDE: Android Studio (Development environment).
3. Frontend: XML with ConstraintLayout & Material Design (UI/UX).
4. Backend: Firebase Auth (User login/registration).
5. Database: Firebase Realtime Database (Storing user roles and data).
6. Cloud Storage: Firebase Storage (File/image management).

8. Block Diagram

```
+-----------------------------+
|  Teacher Mobile             |
|  App                        |
| (Upload Timetable)          |
+-----------------------------+
                |
+-----------------------------+
| (Firebase Realtime)         |
| Stores Lecture Data         |
+-----------------------------+
                |
+-----------------------------+
| Mobile Application          |
| (Reads Timetable &          |
| Generates Reminder)         |
+-----------------------------+
                |
+-----------------------------+
| Students & Teachers         |
| Receive Lecture             |
| Reminder Notification       |
+-----------------------------+
```

9. Diagram


Lecture Timetable Notification App – System Architecture
Teacher (App)
Login
Upload Timetable
Update Lecture/Room
Cancel Lecture
Saves Data
Firebase Backend
Realtime Database
(Stores Timetable)
Fetches Time Data
Automatic Scheduler
(Sends Notification before Lecture Time)
Cloud Functions
(Triggers Notifications)
Student (App)
Login
View Timetable
Receive Notifications
Room & Time Info
Gets Updates & Notifications
Cloud Messaging (FCM)
(Sends Push Notifications)
Sends Notification
Notifications
Teacher & Student
Data Structures at 10:00 AM in Room 204 – Prof. Sharma
DBMS Lab at 11:00 AM in Lab 3
Working Flow
Teacher Uploads Timetable
Data Saved in Firebase
Auto Notification Sent
Students Get Notified!

Thank You !!!